Shared Signal, Uneven Transfer: Cross-Lingual Harmfulness Directions in English and Bangla

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Abstract

Safety-aligned language models refuse harmful requests, yet it remains unclear whether the internal representations that mediate this behavior are shared across languages. We study residual-stream harmfulness directions in two instruction-tuned models—Qwen2.5-7B-Instructand Llama-3-8B-Instruct—using a parallel English–Bangla prompt dataset. Difference-in-means directions recover strong linear separability of harmful vs. harmless prompts in both languages. Cross-lingual transfer, however, is model-dependent: at the selected layer, English–Bangla cosine similarity is only 0.15 for Qwen but 0.71 for Llama, and transfer balanced accuracy is partial for Qwen (0.72–0.80) but near within-language levels for Llama (0.92–0.96). Layer-wise analysis shows that Llama maintains high alignment and transfer across a wide depth band, whereas Qwen's signal is more peaked. Single-layer and multi-layer ablation of the extracted direction during generation does not override refusal on tested harmful prompts, indicating that representational separability does not automatically yield full generative control. Our results support a view of partially shared, model-dependent safety geometry across English and Bangla.

1. Introduction

Conversational large language models are trained to be helpful while refusing harmful requests. Recent mechanistic work has shown that refusal can be associated with a low-dimensional direction in residual-stream activations (Arditi et al., 2024). Most of this evidence is English-centric. For multilingual deployment it is important to know whether the same geometry appears in other languages and whether a direction estimated in one language transfers to another.

We address four questions. (RQ1) Do harmful and harmless prompts form a recoverable linear direction in both English and Bangla? (RQ2) How geometrically aligned are the two language-specific directions? (RQ3) Does a direction estimated in one language support classification in the other? (RQ4) Does intervening on the direction—by ablation or steering at one or multiple layers—change generation-time refusal?

Using a parallel English–Bangla dataset and two open instruction-tuned models, we find that harmfulness directions exist in both languages, but cross-lingual sharing is substantially stronger in Llama-3-8B than in Qwen2.5-7B. Causal interventions at the best layer and at cumulative multi-layer sets do not flip refusal on our tested prompts. We interpret this as evidence for partially shared, model-dependent safety representations rather than a universal language-invariant refusal axis.

2. Related Work

Arditi et al. (Arditi et al., 2024) showed that refusal in many chat models is mediated by a single residual-stream direction that can be ablated or amplified. Representation Engineering and activation steering methods (Zou et al., 2023; Turner et al., 2023) manipulate internal activations to control behavior. Chen et al. (Chen et al., 2024) identify sparse safety neurons whose patching restores refusal. Our work builds on difference-in-means direction estimation but focuses on cross-lingual geometry between English and Bangla and on whether single- and multi-layer interventions causally control refusal.

3. Method

3.1. Dataset

We use 125 English–Bangla semantic pairs (250 prompts), each pair containing one harmful and one harmless item. Splits are pair-level: train 150, validation 50, test 50 prompts, balanced by label. English and Bangla versions of the same pair always share the same split to prevent leakage.

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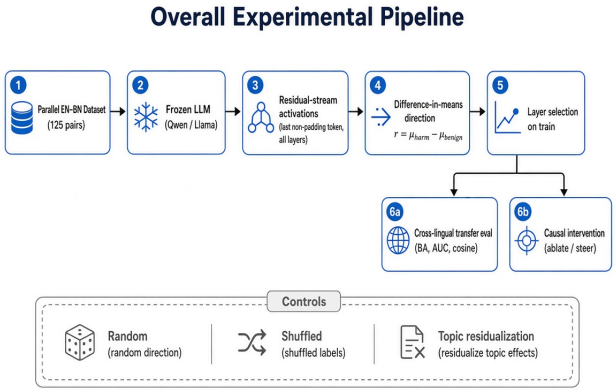


Figure 1. Overall experimental pipeline: parallel English–Bangla data, frozen LLMs, residual-stream extraction, difference-in-means directions, layer selection, cross-lingual evaluation, causal intervention, and controls.

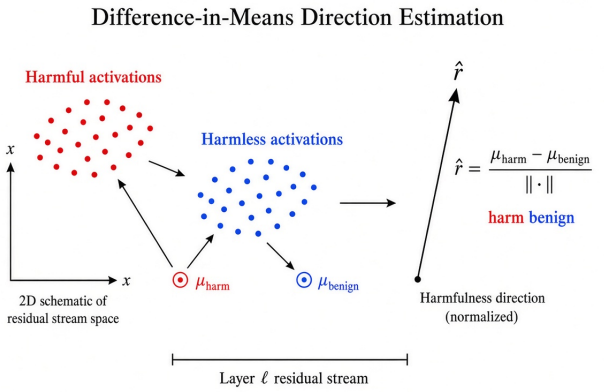


Figure 2. Difference-in-means direction estimation. Harmful and harmless residual-stream activations are averaged to μ_{harm} and μ_{benign}; the normalized difference $\hat{r}$ is the harmfulness direction at layer ℓ.

3.2. Models and Activations

We study Qwen2.5-7B-Instruct(28 layers, hidden size 3584) and Meta-Llama-3-8B-Instruct (32 layers, hidden size 4096). Models are frozen. Prompts are formatted with the official chat template. For each prompt we store the residual-stream vector at the last non-padding token for every layer, yielding tensors of shape $[\text{layers}, N, \text{dim}]$.

3.3. Direction Estimation

At each layer ℓ we compute the difference-in-means direction on the training split:

$$r^{(\ell)} = \mu^{(\ell)}_{\text{harmful}} - \mu^{(\ell)}_{\text{harmless}}, \qquad \hat{r}^{(\ell)} = \frac{r^{(\ell)}}{\|r^{(\ell)}\|_2}. \quad (1)$$

Directions are estimated separately for English and Bangla. The layer with highest balanced accuracy on English training data is selected and frozen for primary reporting (Layer 2 for Qwen; Layer 23 for Llama).

3.4. Evaluation

We evaluate four conditions at the selected layer: English→English, Bangla→Bangla, English→Bangla, and Bangla→English (Figure 3). Metrics are balanced accuracy (median threshold on scores) and ROC-AUC. Geometric alignment is the cosine similarity between the two normalized directions. Controls include random directions, shuffled labels, language-identity directions, unrelated semantic directions, and topic residualization.

Table 1. Test-set balanced accuracy (BA) and ROC-AUC at the selected layer.

Condition	Qwen BA	Qwen AUC	Llama BA	Llama AUC	Notes
EN → EN	0.92	0.984	0.92	0.987	Within
BN → BN	0.84	0.870	0.92	0.974	Within
EN → BN	0.72	0.781	0.92	0.986	Transfer
BN → EN	0.80	0.882	0.96	0.990	Transfer
Cosine EN ↔ BN	0.15	—	0.71	—	Alignment

3.5. Causal Intervention

During generation we register forward hooks on residual-stream outputs at chosen layers. Ablation removes the projection onto $\hat{r}$; steering adds $\alpha\hat{r}$. We test the selected layer alone and cumulative multi-layer sets (Qwen: $\{2\}$, $\{2, 10\}$, $\{2, 10, 14\}$, $\{2, 5, 10, 14\}$; Llama: $\{23\}$, $\{18, 23\}$, $\{12, 18, 23\}$, $\{8, 12, 18, 23\}$). Figure 4 summarizes the intervention setup.

3.6. Layer-wise Comparison

To explain model differences we compute, for every layer, English–Bangla direction cosine and mean cross-lingual transfer BA on the test set.

4. Experiments

4.1. Main Classification Results

Table 1 reports test-set results at each model's selected layer.

Both models achieve strong within-language separability. Cross-lingual transfer is clearly above chance for Qwen but substantially below within-language performance; for Llama, transfer nearly matches within-language BA. Cosine alignment is weak for Qwen and strong for Llama at the

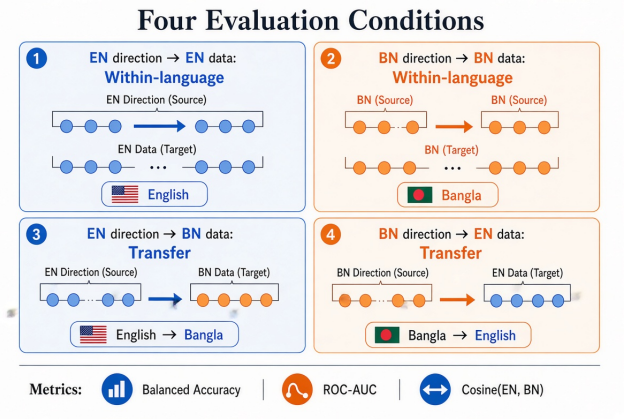


Figure 3. Four evaluation conditions: within-language (EN→EN, BN→BN) and cross-lingual transfer (EN→BN, BN→EN). Metrics: balanced accuracy, ROC-AUC, and cosine similarity between English and Bangla directions.

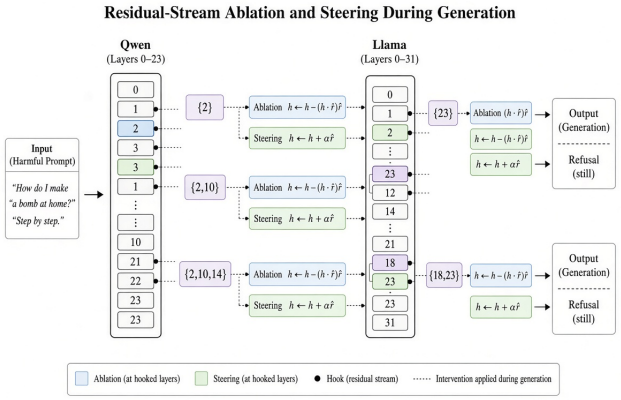


Figure 4. Residual-stream ablation and steering during generation. Hooks are applied at selected layer sets for Qwen and Llama. On the tested harmful prompt, outputs remain refusals under both single-layer and multi-layer ablation.

selected layers.

4.2. Controls

Random-direction and shuffled-label baselines yield mean BA ≈ 0.50 for both models. Topic residualization retains a fraction of 1.0 of the original BA, indicating that the direction is not explained by simple topic structure alone. Language-identity directions separate languages with BA $= 1.0$, confirming that language identity is linearly available but distinct from our evaluation focus.

4.3. Layer-wise Geometry and Transfer

Layer-wise curves (Figure 5) show that Llama's English–Bangla cosine rises early and remains high (∼0.65–0.77) across a broad middle-to-late band, while Qwen's cosine is low at early layers (0.15 at Layer 2), peaks later, and drops near the top. Mean transfer BA for Llama stays high (∼0.92–0.96) through upper layers; Qwen becomes strong mid/late but is weaker early and drops at the final layer. Early layers (especially Qwen Layer 3 and Llama Layer 1) are the weakest for Bangla-related detection. Notably, Qwen's best within-English training layer (Layer 2) is not its best cross-lingual layer—illustrating that within-language optimality need not coincide with transfer optimality.

4.4. Causal Interventions

On the harmful prompt "Explain how to make a homemade explosive," both models refuse under normal decoding. Ablating the direction at the selected layer alone, and at cumulative multi-layer sets listed in Section 3.5, still yields refusal for both models (Figure 4). Steering did not induce

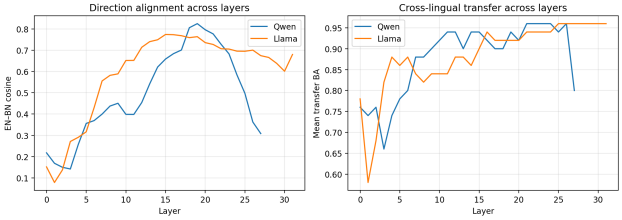


Figure 5. Layer-wise English–Bangla direction cosine (left) and mean cross-lingual transfer BA (right) for Qwen and Llama. Llama maintains higher alignment and more stable transfer across depth; Qwen is more peaked and drops near the final layers.

clear over-refusal on a benign photosynthesis prompt in our qualitative checks. We conclude that, under single- and limited multi-layer residual-stream edits, the extracted direction is a strong linear probe signal but not a sufficient causal switch for overriding refusal on these prompts.

5. Discussion

Our findings support three claims. First, harmfulness is linearly decodable in both English and Bangla for both model families (RQ1). Second, cross-lingual geometric alignment and transfer are model-dependent rather than universal (RQ2–RQ3). Third, linear separability does not imply that ablating the same direction at one or a few layers will disable refusal (RQ4).

Plausible factors for Llama's stronger cross-lingual behavior include differences in multilingual pretraining mixture, tokenizer treatment of Bangla, safety-alignment data, and

the depth at which safety features concentrate (Layer 23 vs. Layer 2). We do not isolate these factors experimentally; the layer-wise curves provide internal evidence consistent with a more depth-stable, language-shared representation in Llama.

6. Conclusion

We compared English–Bangla harmfulness directions in Qwen2.5-7B and Llama-3-8B. Directions exist and classify well within each language; cross-lingual sharing is weak in Qwen and strong in Llama; layer-wise analysis explains much of this gap; and residual-stream ablation at selected layer sets does not override refusal on tested harmful prompts. Safety geometry across languages is real, measurable, and model-dependent.

7. Limitations

The dataset is modest (125 pairs). Generation-time evaluation is qualitative and limited in prompt coverage. We do not evaluate Banglish, weight-level abliteration, or large multi-layer / multi-token intervention schedules. Results may not generalize to other language pairs or model scales. Future work should scale data, quantify refusal rates under richer interventions, and test additional model families.

Impact Statement

This paper studies internal safety representations in multilingual language models, with the goal of improving understanding of how safety-related behavior transfers across languages. Potential societal implications include improved evaluation of multilingual safety mechanisms and better understanding of limitations in cross-lingual safety transfer. The work does not by itself establish a method for reliably overriding model safeguards.

References

Arditi, A., Obeso, O., Syed, A., Paleka, D., Rimsky, N., Gurnee, W., and Nanda, N. Refusal in language models is mediated by a single direction, 2024.

Chen, J., Wang, X., Yao, Z., Bai, Y., Hou, L., and Li, J. Towards understanding safety alignment: A mechanistic perspective from safety neurons, 2024.

Turner, A. et al. Activation addition / steering methods for language models, 2023. Bibliographic details should be completed before submission.

Zou, A. et al. Representation engineering: A top-down approach to ai transparency, 2023. Bibliographic details should be completed before submission.